

VC02 Deployment Environment

VM Setup, IP Allocation & Sprint Schedule

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What We'll Cover

01

Deployment Overview

Architecture, network, and environment structure

02

Group & VM Allocation

12 groups · 36 VMs · IP addresses on 172.16.16.0/24

03

Student Access Guide

SSH, Git, deploy workflow, and credentials

04

Sprint Schedule

Week 1 draft demo → Sprint reviews → Final demo

Environment Architecture — 12 Groups × 3 VMs Each

Hypervisor	Proxmox VE / VMware
Network	172.16.16.0/24 (flat)
Total VMs	36 VMs (12 × 3)
Public Gateway	Nginx reverse proxy
OS per VM	Ubuntu 24.04 LTS
Git Platform	GitHub (12 repos)

FE

Frontend VM

IP range: 172.16.16.189—.200

OS: Ubuntu 24.04 . Username : pnc . Password : 1234567

BE

Backend VM

IP range: 172.16.16.177—.188

OS: Ubuntu 24.04 . Username : pnc . Password : 1234567

DB

Database VM

IP range: 172.16.16.201—.212

OS: Ubuntu 24.04 . Username : pnc . Password : 1234567

Group & IP Reference — Groups 1–6

Group	Backend VM	Frontend VM	DB VM	Domain
Group 1	172.16.16.177	172.16.16.189	172.16.16.201	group1.pnc.local
Group 2	172.16.16.178	172.16.16.190	172.16.16.202	group2.pnc.local
Group 3	172.16.16.179	172.16.16.191	172.16.16.203	group3.pnc.local
Group 4	172.16.16.180	172.16.16.192	172.16.16.204	group4.pnc.local
Group 5	172.16.16.181	172.16.16.193	172.16.16.205	group5.pnc.local
Group 6	172.16.16.182	172.16.16.194	172.16.16.206	group6.pnc.local

Backend .177–.188

Frontend .189–.200

DB .201–.212

Group & IP Reference — Groups 7–12

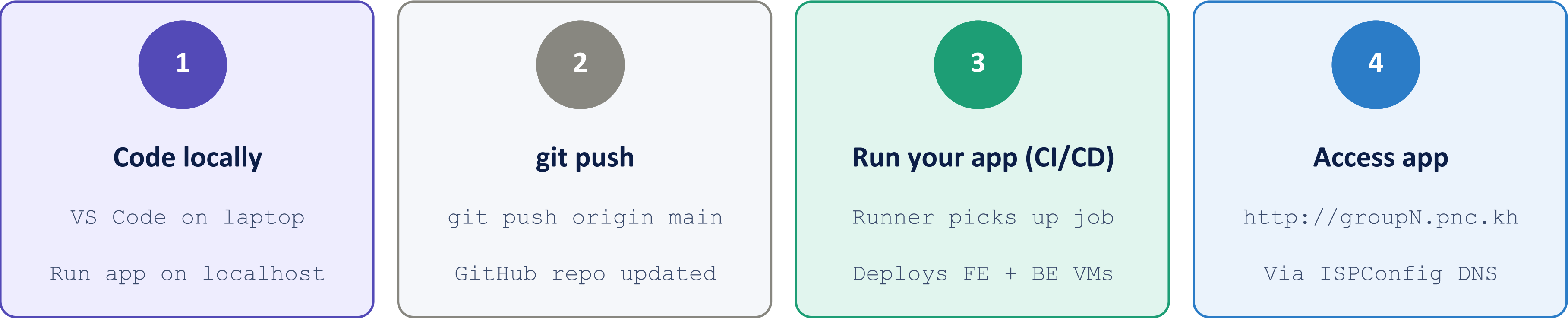
Group	Backend VM	Frontend VM	DB VM	Domain
Group 7	172.16.16.183	172.16.16.195	172.16.16.207	group7.pnc.local
Group 8	172.16.16.184	172.16.16.196	172.16.16.208	group8.pnc.local
Group 9	172.16.16.185	172.16.16.197	172.16.16.209	group9.pnc.local
Group 10	172.16.16.186	172.16.16.198	172.16.16.210	group10.pnc.local
Group 11	172.16.16.187	172.16.16.199	172.16.16.211	group11.pnc.local
Group 12	172.16.16.188	172.16.16.200	172.16.16.212	group12.pnc.local

Backend .177–.188

Frontend .189–.200

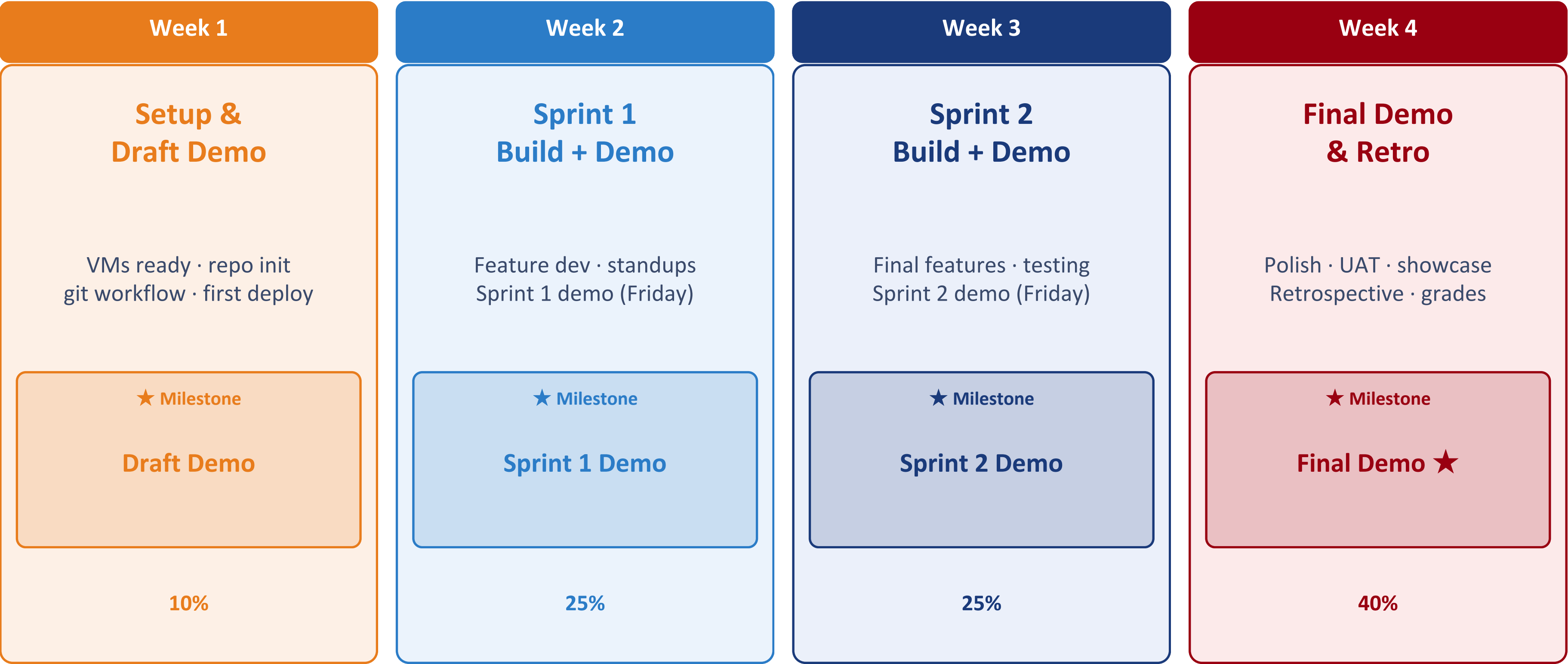
DB .201–.212

Option B — Local Dev + Git Push



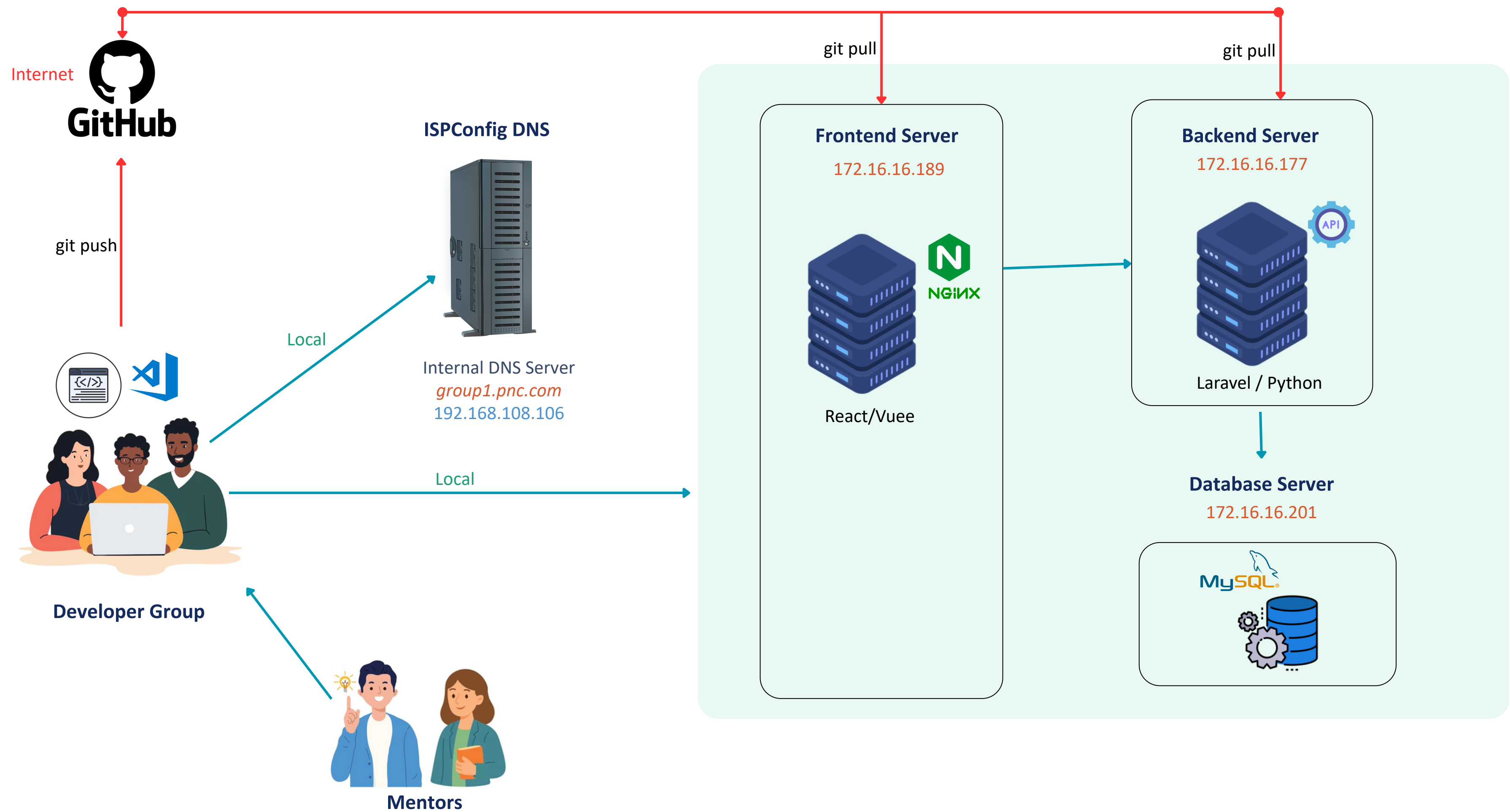
Frontend VM	172.16.16.189 (group 1)	Nginx port 80 · app port 3000
Backend VM	172.16.16.177 (group 1)	API server port 8000
Database VM	172.16.16.201 (group 1)	MySQL port 3306 · internal only

4-Week Deployment & Review Timeline



★ Build + Review happen in the SAME week — ship working code every Friday.

INFRASTRUCTURE DIAGRAM



EVERY SPRINT

Deploy Early. Demo Often.

Every sprint ends with a working product on your VM — not code that "works locally."